

Advancing Basin-Scale Groundwater Storage Estimation Through Hybrid Time-Series Imputation, Spatiotemporal Interpolation, and Satellite-Derived Leakage Correction

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May 2026

1. Introduction

1.1 Overall Objective and Significance

Groundwater sustains roughly half of global irrigation and a third of municipal supply, yet remains the most poorly monitored component of the terrestrial water budget (Scanlon et al., 2023). In arid and semi-arid basins, where surface reservoirs are increasingly stressed by warming-driven evaporation and growing demand, reliable quantification of groundwater storage change is essential for drought preparedness, allocation policy, and long-term sustainability assessment. The problem is acute: Jasechko et al. (2024) analyzed 170,000 wells across 1,693 aquifer systems worldwide and found rapid declines exceeding 0.5 m/yr in a substantial fraction, with 30% of regional aquifers showing accelerating loss over the past four decades.

Three independent lines of evidence bear on basin-scale groundwater storage change, each with fundamental limitations. In situ monitoring well networks provide direct water-level observations but are temporally sparse and spatially uneven. Public databases such as the USGS National Water Information System contain millions of individual measurements, yet the median well has fewer than 50 observations spanning decades of record, with gaps of multiple years commonplace. Satellite gravimetry from the Gravity Recovery and Climate Experiment (GRACE) and its successor GRACE-FO resolves total water storage anomalies globally at monthly resolution (Tapley et al., 2004), but the native spatial resolution of approximately 300 km makes basin-scale interpretation subject to signal leakage across boundaries (Vishwakarma et al., 2018), and isolating the groundwater component requires subtracting auxiliary estimates of soil moisture, snow, surface water, and canopy storage from land-surface models whose own fidelity is uncertain. Land-surface models such as those within the Global Land Data Assimilation System (GLDAS; Rodell et al., 2004) provide spatially continuous, temporally complete estimates but inherit assumptions about subsurface representation, irrigation withdrawals, and terminal-lake hydrology that may not hold in heterogeneous, anthropogenically stressed basins.

The overall objective of this dissertation is to develop and demonstrate an integrated workflow that substantially improves basin-scale groundwater storage estimation in heterogeneous basins by: (1) reconstructing temporally complete well records from sparse in situ observations using a novel hybrid imputation framework; (2) interpolating those reconstructed records to produce continuous spatiotemporal groundwater-level fields; and (3) using those fields to derive spatially distributed GRACE leakage corrections that replace the basin-uniform scalars currently applied in practice. The Great Salt Lake Basin (GSLB), a closed hydrologic system in the Intermountain West with concentrated anthropogenic pumping along the Wasatch Front and a hydrologically significant terminal lake, serves as the demonstration basin throughout.

1.2 Background

Groundwater Monitoring and the Gap Problem

Evans et al. (2020a) developed the Groundwater Level Mapping Tool (GWLMT), an open-source web application that couples Extreme Learning Machines (ELM; Huang et al., 2006) with Earth observation data from GRACE and GLDAS to impute missing groundwater levels. Evans et al. (2020b) showed that the ELM-based approach achieves superior imputation compared to conventional interpolation. Ramirez et al. (2022) extended this framework by incorporating inductive bias from remote sensing, and Ramirez et al. (2023) further advanced the methodology through iterative refinement exploiting both spatial and temporal correlations. Stevens et al. (2025) and Shepard et al. (2025) applied these methods to basin-scale storage assessment in the Central Valley and Klamath watershed, respectively.

These ELM-based approaches operate primarily in the temporal domain: each well is imputed independently using auxiliary time series as features. Jeong et al. (2020) demonstrated that recurrent neural networks can reconstruct missing groundwater levels with high accuracy for wells with correlated neighbors, but performance degrades sharply beyond one to two year gaps. Wunsch et al. (2021) compared LSTM, CNN, and NARX architectures, finding that multivariate temporal models outperform univariate ones but remain limited by training data availability during gaps. Gharehbaghi et al. (2022) and Lin et al. (2022) showed GRU networks perform comparably to LSTM. However, all these

approaches treat wells independently, incorporating spatial information only through satellite-derived covariates rather than directly exploiting the cross-well correlation structure.

Regional correlation-based approaches represent an intermediate strategy. Levy et al. (2025) developed ARCHI (Automated Regional Correlation Analysis for Hydrologic Record Imputation), a USGS R package that imputes missing data in target records by linear regression using more complete reference records as predictors. ARCHI's iterative algorithm allows each site to serve as both target and reference, progressively growing the pool of complete records. While ARCHI effectively exploits spatial correlation through donor regression, it operates within a linear framework and does not incorporate auxiliary climatic forcings or nonlinear temporal dynamics.

Matrix completion offers a fundamentally different perspective by arranging all well records as a partially observed matrix and exploiting low-rank structure. Candes and Recht (2009) proved that low-rank matrices can be recovered from surprisingly few observed entries. Sharma, Kim, and Tayerani Charmchi (2024) evaluated SoftImpute for monthly groundwater levels, finding it excels in sparse networks -- but treated it as a standalone method without temporal coupling.

Hasani et al. (2021) introduced Liquid Time-Constant Networks, and the subsequent Closed-form Continuous-depth architecture (Hasani et al., 2022) achieves one order of magnitude faster training by eliminating numerical ODE solvers. No prior work has combined matrix completion with continuous-time neural networks for groundwater imputation. This dissertation proposes a coupled MC+LNN framework: matrix completion exploits cross-well correlations while the LNN refines using continuous-time dynamics conditioned on auxiliary climatic forcings.

Spatial Interpolation of Groundwater Levels

Geostatistical methods, particularly kriging, have been dominant but their stationarity assumptions are frequently violated in heterogeneous basins. Empirical Orthogonal Function (EOF) analysis decomposes spatiotemporal fields into temporal modes and spatial loadings, enabling interpolation by estimating loadings at unobserved locations rather than raw values directly.

GRACE Leakage Correction

Long et al. (2014) established forward modeling for GRACE leakage correction. Ma et al. (2024) introduced Coordinated Forward Modeling at 0.5-degree scale. Tripathi et al. (2022) showed basin-average corrections can be misleading. Li et al. (2024) estimated pixel-scale leakage factors using in situ data.

Great Salt Lake Basin

The GSLB covers approximately 93,000 km². Consumptive uses have depleted inflows by 39% (Null and Wurtsbaugh, 2020). Wine (2019) argued consumptive use dominates over climate change. Bigalke et al. (2025) attributed the 2022 record low to reduced streamflow and warming. Hall et al. (2024) documented 68.7 km³ groundwater loss across the Great Basin. Zamora and Inkenbrandt (2024) revised the groundwater contribution to the GSL upward to 10% of total inflows.

1.3 Specific Objectives

Objective 1 (Paper 1). Quantify multi-decadal groundwater storage change in the GSLB (2002-2024) by integrating GRACE-derived, GLDAS-derived, and in situ estimates, and assess the consequences of surface-water inclusion and GRACE leakage correction.

Objective 2 (Paper 2). Develop and validate a hybrid imputation framework combining PCHIP for short gaps with coupled Matrix Completion and Liquid Neural Network (MC+LNN) for long gaps, using auxiliary climatic forcings as continuous-time inputs.

Objective 3 (Paper 3). Develop a spatiotemporal interpolation framework using spatial trend decomposition and EOF analysis to produce continuous groundwater-level fields, and apply those fields to derive spatially distributed GRACE leakage correction factors.

2. Objective 1 (Paper 1)

2.1 Objective

To quantify groundwater storage change in the GSLB from 2002 through 2024 using multiple independent methods, evaluate methodological sensitivities, and produce a continuous record for water-resources planning.

2.2 Background

Sustained low Great Salt Lake levels have raised ecological concerns (Abbott et al., 2023). Prior GRACE studies addressed the broader Great Basin (Hall et al., 2024) and localized loss via GPS (Young, Kreemer, and Blewitt, 2021), but no multi-method record specific to the full GSLB has been produced.

2.3 Methods

Five GWSa estimates were computed: GRACE-raw, GRACE-sw (surface-water adjusted), GRACE-Lf (leakage-corrected, $Lf=2$), GLDAS-2.2, and GWDM (in situ via the GWLMT workflow of Evans et al., 2020a; Ramirez et al., 2023).

2.4 Results

All methods identified two major drawdowns (2012-2016, 2019-2022). GRACE-Lf yielded 10.1 km³ loss for 2011-2016, consistent with 10.9 ± 2.8 km³ from GPS. Surface water accounts for 31% of storage change. The leakage factor improved GRACE-in situ correlation from 0.17 to 0.77.

3. Objective 2 (Paper 2)

3.1 Objective

To develop, validate, and benchmark a hybrid imputation framework combining classical interpolation, matrix completion, and continuous-time neural networks for groundwater-level time series with multi-year gaps.

3.2 Methods

Stage 1: PCHIP. Gaps ≤ 24 months are filled using shape-preserving PCHIP interpolation, densifying the network for better donor correlations in Stage 2.

Stage 2: MC+LNN. Large gaps are filled in three phases: (a) ARCHI donor regression selects top-15 correlated wells and provides trend-aware initialization; (b) SoftImpute matrix completion on a composite matrix of target, donors, GLDAS auxiliary variables, and seasonal encoding; (c) LNN CFC temporal refinement using continuous-time dynamics:

$$x(t + dt) = x \cdot e^{-\lambda \cdot dt} + (b / \lambda) \cdot (1 - e^{-\lambda \cdot dt})$$

MC predictions serve as reservoir input during gaps, but readout trains only on real observations. Validated on 592 GSLB wells (2000-2023) with random missing (5-50%) and consecutive gap (1-5 year) cross-validation.

3.3 Anticipated Results

Scenario	KGE	R ²	RMSE (ft)
5% random missing	0.837 ± 0.085	0.771	2.53
20% random missing	0.853 ± 0.051	0.787	2.64
50% random missing	0.847 ± 0.035	0.788	2.74
1-year consecutive gap	0.783 ± 0.116	0.703	2.98
3-year consecutive gap	0.802 ± 0.076	0.730	3.02
5-year consecutive gap	0.815 ± 0.063	0.744	2.91

PCHIP small-gap fill is critical: replacing it with LNN degrades KGE by 10-19%. The framework imputes all 592 wells to completeness.

4. Objective 3 (Paper 3)

4.1 Objective

To produce continuous groundwater-level fields via spatiotemporal interpolation and derive spatially distributed GRACE leakage correction factors.

4.2 Methods

Three-stage interpolation: (1) degree-2 polynomial trend surface of WTE mean as $f(\text{lat}, \text{lon}, \text{elevation})$, $R^2=0.957$; (2) EOF decomposition of detrended residuals via SVD; (3) IDW interpolation of spatial loadings to grid cells. The interpolated fields are converted to storage anomalies and used to calibrate pixel-wise leakage factors $L_f(\phi, \lambda)$, benchmarked against mascon-based GWSa (Watkins et al., 2015).

4.3 Anticipated Results

EOF interpolation achieves RMSE 32.3 ft (49% reduction vs IDW, 73% vs kriging). The pixel-wise L_f grid is expected to show largest factors along the Wasatch Front, with smaller values in the West Desert.

5. Timeline

Period	Activity
May 2026	Paper 1 published; Paper 2 validation complete
Jun-Aug 2026	Paper 2 manuscript drafting
Sep 2026	Paper 2 submission
Sep-Dec 2026	Paper 3 gridded L_f calibration and sensitivity analyses
Jan-Feb 2027	Paper 3 results, mascon benchmarking
Mar 2027	Paper 3 manuscript drafting
Apr 2027	Paper 3 submission; dissertation compilation

Period	Activity
May-Jun 2027	Defense and graduation

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